SANJANA RANI

Graduate Aerospace Engineer

sanjanakamboj98@gmail.com | +91-9996673079 | linkedin.com/in/sanjana-rani

PROFESSIONAL SUMMARY
Aerospace engineer with hands-on CATIA V5 and Creo experience, including 3D modelling, 2D engineering drawings, and ISO GPS tolerancing for test rig assemblies. Complements design expertise with strong Python scripting skills for workflow automation and data processing with exposure to structural analysis (ABAQUS), CFD, and experimental setups.

Key Skills

Design & Simulation: CATIA-V5, Creo, SolidWorks, ANSYS Fluent, ABAQUS, Paraview
Data & Programming: Python, MATLAB
Experimental & Testing: Wind-tunnel Testing, PIV (including Stereo-PIV), Flow Visualization
Languages: English (Fluent), German (A1, improving), French (Beginner)

Professional Experience

DLR Institute of Electrified Aero Engines (DLR e.V.), Cottbus, Germany *Feb 2024 - Sep 2024*
Graduate Research Assistant

- Coordinated the development, integration, and experimental validation for EMC testing of acoustic sensors as part of the LU(FT)^2 2030 project, liaising across engineering sub-teams and tracking milestone completion.
- Designed and executed Python-based signal processing pipelines (FFT, coherence, spectral subtraction) to assess sensor performance, measurement integrity, and noise susceptibility under varying operational conditions.

Chair of Aerodynamics and Fluid Mechanics, BTU, Cottbus, Germany *Aug 2023 – Feb 2024*
Student Assistant

- Assisted with mechanical assembly, experimental setup, and flow-quality checks for the TEKUS / DEPIK zero-gravity convection project, ensuring alignment across setup, testing, and validation phases.
- Created 2D engineering drawings of test rig components and full assembly using Creo, applying ISO GPS tolerancing standards based on original part dimensions to ensure dimensional accuracy and fit.
- Conducted PIV and optical flow visualisation to analyse velocity fields, flow stability, and laminarity in experimental aerodynamic configurations.

Education

Joint MSc in Transfer of Fluid Materials in Aeronautical & Space Applications (TFM-ASA) *Sep 2022 - Sep 2024*
Université de Bordeaux (Bordeaux, France) Université Catholique de Louvain (Louvain-la-Neuve, Belgium) Brandenburgische Technische Universität (Cottbus, Germany)

- **Thesis:** Electromagnetic Interference and its effects on Acoustic Sensor measurements
- **Selected Coursework:** Experiments in Aerodynamics and Fluid Mechanics, Flow Measurement Techniques, Gasdynamics and Reacting Flows, Thermodynamics of Irreversible Phenomena.

B.E. in Aerospace Engineering *Aug 2018 - Jun 2022*
Chandigarh University, Gharaun, India

- **Core Coursework:** Aerodynamics, Wind Tunnel Testing, Flight Mechanics, Aircraft Propulsion, CFD, Rocket Propulsion
- **Final Project:** CFD Analysis of a hypersonic nozzle using ANSYS Fluent

Selected Projects

Brahmastra Research & Development Center — Freelance Project Trainee *Apr 2023 – May 2023*

- Designed and modeled a pintle injector in CATIA V5, developing parametric 3D geometry for rocket propulsion applications.
- Built an integrated CAD-to-CFD workflow linking CATIA V5 geometry with ANSYS Fluent simulations, automating the handoff between design and analysis phases.

Turbine Stage Design Analysis — Université Catholique de Louvain *2023*

- Performed 1D mean-line design of turbine stage given certain input boundary conditions to compute thermodynamic states and velocity triangles, followed by 2D aerodynamic design of a rotor blade passage.
- Implemented blade geometry using Parablade, meshed the flow domain in GMSH, and conducted steady 2D compressible RANS simulations in SU2.
- Post-processed aerodynamic performance using Python and ParaView, evaluating velocity, Mach number, and efficiency-related metrics.